

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	Saee B	Reg. No.:	192524023
Section / Batch:	AI and PS / 2025	Date:	20/07/2026

Code of Conduct

I Saee B (192524023) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. S

Instructions

- This test contains 5 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
GRAND TOTAL:			100 Marks

Annexure A – Analytical Problem Solving

S.No	REG NO	NAME	ANALYTICAL PROBLEM SOLVING
1	192472062	ANTHOTI ROHITH	1. Perform the following operations in LIFO data structure having an array size of 5. Push(45), Push(22), Push(11), Push(77), Pop(), Push(44), Push(33), Pop(), Pop(), Pop(). Print the peak element of the data structure and its index position. With a neat sketch explained in detail. 2. How do you implement Queue data structure using Single Linked List? With a neat diagram discussing Enqueue and Dequeue operations using SLL.
2	192572047	AARTHI SREE G	1. Perform the following operations in Circular Queue having a size of 5. Enqueue(20), Enqueue(10), Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(90), Enqueue(100), Dequeue(), Dequeue(), Enqueue(60), Enqueue(90). Finally display the front and rear elements with its positions. Discuss in detail. 2. Implement Binary searching Algorithm for searching an element and its position having the following 10 elements. Elements are 66, 111, 212, 323, 432, 543, 643, 649, 777, 888. Case 1: Key element = 111, Case 2: Key element=888, Case 3: Key element = 999. Analyse the concept in detail.
3	192525192	ARAVIND S	1. Pictorially insert an element at 3rd position in a doubly linked list, having 6 elements in the list and delete the element at 2nd position from the list. Implement and analyse it. 2. Perform the following operations using stack. Assume the size of the stack is 5 and have a value of 22, 55, 33, 66, 88 in the stack from 0 position to size-1. Now perform the following operations: 1) Invert the elements in the stack, 2) Pop(), 3) Pop(), 3) Pop(), 4) Push(90), 5) Push(36), 6) Push(11), 7) Push(88), 8) Pop(), 9) Pop(). Draw the diagram of the stack and illustrate the above operations and identify where the top is?
4	192571054	CHALAMALA MOKSHITHA REDDY	1. Convert the following infix expression into a post fix expression. $B - (C/D + (E \% F * G) / H) * J$. Also, evaluate the given postfix expressions. a) 9 3 8 4 /*+ b) 6 2 + 6 3 2. Illustrate the queue operation using following function calls of size = 5. Enqueue(21), Enqueue(67), Enqueue(89), Dequeue(), Enqueue(15), Enqueue(30), Enqueue(52), Dequeue(), Dequeue(), Dequeue(), Dequeue().

5	192511465	DHIVYADHARSAN MS	1. Identify if the given expressions are balanced or not. (i) $[9 * (k+8 - \{ 6 + e \} + 77) / 100]$, (ii) $((a*b) + \{u/2\} - d)$. 2. Given a stack with repeated elements, design an approach to compress it (value & frequency) while maintaining stack operations. Apply stack ADT modifications. Analyze trade-off between time and space
6	192521343	GOPIKA A	1. Two algorithms solve the same problem. Algorithm A uses recursion, while Algorithm B uses an explicit stack. Compare both approaches in terms of memory usage, execution flow, and risk of stack overflow. 2. A stack has a capacity of 5 elements. Explain what happens if the following operations are executed: Push(A), Push(B), Push(C), Push(D), Push(E), Push(F) Identify the error and suggest two possible solutions.
7	192524020	GULLAPALLI SANTOSH KUMAR	1. Insert the following elements 100, 200, 300, 400, 500 into the linked stack and pop the elements 100 and 200 from the linked stack. Show the linked structure before and after linked stack operations. 2. A circular queue has a size of 6. Execute the following operations: Enqueue(10), Enqueue(20), Enqueue(30), Dequeue(), Enqueue(40), Enqueue(50), Enqueue(60), Enqueue(70). Draw the queue after each operation and identify the values of the front and rear pointers.
8	192511397	H SARAVANA	1. Compare the performance of a linear queue and a circular queue when 100 enqueue and dequeue operations are performed alternately. 2. A queue implemented using an array report "Queue Full" even though some positions are empty. a. Analyze why this occurs. b. Explain how a circular queue solves this problem.
9	192525345	INTURI JASWANTH	1. Consider the singly linked list: $10 \rightarrow 20 \rightarrow 30 \rightarrow 40 \rightarrow 50$ Delete the node containing 30. Explain how the links change after deletion. 2. Given two sorted linked lists: L1: $1 \rightarrow 3 \rightarrow 5 \rightarrow 7$ L2: $2 \rightarrow 4 \rightarrow 6 \rightarrow 8$
10	192525211	JUJARU MEHER KOUSHIK	1. Convert the following infix expression into postfix notation using a stack. Show the contents of the stack after processing each symbol. $(A+B)*(C-D)/E$ 2. Convert the following infix expression into postfix form. Explain how operator precedence and associativity influence the conversion. $A+B*C-(D/E+F)*G$
11	192511070	K SUTHESIKA	1. Insert the following elements 150, 250, 350, 450, 550 into the double linked list and delete the elements 150 and 250 from the double linked list. Show the linked structure using a node diagram before and after delete operations.

			2. Mr. XYZ booked a ticket on Pandian Express. He reached the station and was looking for his PNR number in the chart. Call out binary search procedure to search the PNR number 4056786. The ticket numbers are 4056779, 4056780, 4056781, 4056782, 4056783, 4056784, 4056785, 4056786, 4056787, 4056788, 4056789.
12	192472154	KARNATI VIGNESH	1. Convert the expression $((a + b) + c * (d + e) + f) * (g + h)$ to 1) Prefix expression, 2) Postfix expression. 2. Two students obtain different postfix expressions for the same infix expression. Infix: $(A+B)*(C-D)/E$ Student A: $AB+CD-*E/$ Student B: $AB+C*D-E/$ Determine which postfix expression is correct and justify your answer by tracing the conversion process.
13	192521157	KABILAN K	1. How a queue may be maintained by a circular array with $MAXSIZE = 6$. Observe that queue always occupies consecutive locations except when it occupies locations at the beginning and at the end of the array. If the queue is viewed as a circular array, this means that it still occupies consecutive locations. The queue is empty when $count = 0$; Perform the following Enqueue and Dequeue operations. a) Initially QUEUE is empty, b) A, B, C are Enqueued, c) A is Dequeued, d) D, E, F are Enqueued, e) B & C are Dequeued, f) G is Enqueued, g) D & E are Dequeued, h) H & I are Enqueued, i) F is dequeued, j) G & H are dequeued k) I is Dequeued. Demonstrate with a neat sketch of Front, Rear and count in a queue and circular queue. 2. Show the linked list configuration resulting from each series of list operations using the List ADT. Assume that lists L1 and L2 are empty at the beginning of each series. Show where the current position is in the list. (a) $L1.append(10); L1.append(20); L1.append(15);$ (b) $L2.append(10); L2.append(20); L2.append(15); L2.moveToStart(); L2.insert(39); L2.next(); L2.insert(12);$
14	192572105	KATTAMURI VENKATA SRINIVASAGNANES WAR KUMAR	1. Evaluate the following postfix expression using a stack. Show the stack contents after every operation. $23*54*+9-$ 2. Design an algorithm to evaluate any postfix expression using a stack. Explain why stacks are more suitable than queues for postfix evaluation.
15	192525040	KONIJETI KARTHIK	1. Given a singly linked list L: $L1 \rightarrow L2 \rightarrow L3 \dots \rightarrow L_{n-1} \rightarrow L_n$, reorder and print it to: $L1 \rightarrow L_n \rightarrow L2 \rightarrow L_{n-1}$. 2. Insert the following elements 150, 250, 350, 450, 550 into the double linked list and delete the elements 150 and 250 from the double linked list. Show the linked structure using a node diagram before and after delete operations.

16	192521194	LOKESH WARAN T	1. Consider a linked list representing the sales of different products in a store for a week. Each node of the linked list contains the product ID and the number of units sold on a specific day. Implement a linked list data structure and provide functions to: Insert new sales data for a given product on a specific day. Calculate the total sales for each product at the end of the week. Find the product with the highest sales and its corresponding day. Display the sales data for a specific product for the entire week. Use your own data for product. 2. A calculator application receives the following expression: $(8+2)*(6-4)/2$
17	192372296	MANGADHODDI DEEPIKA	1. A hospital treats patients on a First-Come, First-Served (FCFS) basis. Patients arrive in the following order: P1 (General) P2 (Emergency) P3 (General) P4 (General) P5 (Emergency) <i>Questions:</i> ➤ If a normal queue is used, what will be the treatment order? ➤ Explain why this approach is not suitable. ➤ Suggest a better queue-based solution and justify your answer 2. A calculator application receives the following expression: $(10+5) * (3-4)/2$
18	192513069	MADHUMITHA K	1. Find the difference between the minimum and maximum values in a single linked list having 34->77->22->55->66->11-> in the list. 2. The double linked list has two links in each node as "prev", "data" and "next". Display the elements from DLL in forward and reverse order and find the average of those 'n' elements.
19	192525318	MADINENI DHANUSH KUMAR	1. Print all the odd and even counts present in the single linked list having 10 elements in the list 77->88->22->11->66->27->55->65->18->10-> 2. Search linearly a key elements K=20 present in the list of single lined list. If so print "YES" else print "NO".
20	192571007	MURSHIDHA M	1. A supermarket has only one billing counter. Customers arrive in the order: C1, C2, C3, C4, C5. After C2 is served, a senior citizen arrives. <i>Questions:</i> ➤ Explain how the queue changes if a normal queue is used. ➤ Design a queue-based solution that prioritizes senior citizens while maintaining fairness. 2. Assume a single character is present in the data field of a Single Linked List with 6 nodes as S-> I-> M-> A-> T-> S. Write a

			function to print the same in reverse order using stack data structure.
21	192525063	NALIKIRI SIVA VENKATA TRIVIKRAM	1. Write an algorithm called Find-Largest that finds the largest number in an array using a divide-and-conquer strategy. 2. Analyze how to print all the prime integers present in the Single linked list having a list of 24->21->57->31->12->47->7->38->.
22	192511294	NIRUTHIKA SRI PRAKASH	1. Perform the following operations in LIFO data structure having an array size of 5. Push(45), Push(22), Push(11), Push(77), Pop(), Push(44), Push(33), Pop(), Pop(), Pop(). Print the peak element of the data structure and its index position. With a neat sketch explained in detail. 2. How do you implement Queue data structure using Single Linked List? With a neat diagram discussing Enqueue and Dequeue operations using SLL.
23	192525039	OSURU SIVA CHARAN	1. Perform the following operations in Circular Queue having a size of 5. Enqueue(20), Enqueue(10), Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(90), Enqueue(100), Dequeue(), Dequeue(), Enqueue(60), Enqueue(90). Finally display the front and rear elements with its positions. Discuss in detail. 2. Implement Binary searching Algorithm for searching an element and its position having the following 10 elements. Elements are 66, 111, 212, 323, 432, 543, 643, 649, 777, 888. Case 1: Key element = 111, Case 2: Key element=888, Case 3: Key element = 999. Analyse the concept in detail.
24	192572014	PRITHAM G	1. Pictorially insert an element at 3rd position in a doubly linked list, having 6 elements in the list and delete the element at 2nd position from the list. Implement and analyse it. 2. Perform the following operations using stack. Assume the size of the stack is 5 and have a value of 22, 55, 33, 66, 88 in the stack from 0 position to size-1. Now perform the following operations: 1) Invert the elements in the stack, 2) Pop(), 3) Pop(), 3) Pop(), 4) Push(90), 5) Push(36), 6) Push(11), 7) Push(88), 8) Pop(), 9) Pop(). Draw the diagram of the stack and illustrate the above operations and identify where the top is?
25	192525195	R BHAVYA	1. Convert the following infix expression into a post fix expression. $B - (C/D + (E \% F * G) / H) * J$. Also, evaluate the given postfix expressions. a) 9 3 8 4 /*+ b) 6 2 + 6 3 2. Illustrate the queue operation using following function calls of size = 5. Enqueue(21), Enqueue(67), Enqueue(89), Dequeue(), Enqueue(15), Enqueue(30), Enqueue(52), Dequeue(), Dequeue(), Dequeue(), Dequeue(), Dequeue().

26	192524066	RAGUL M	1. Identify if the given expressions are balanced or not. (i) $[9 * (k+8 - \{ 6 + e \} + 77) / 100]$, (ii) $((a*b) + [u/2] - d)$. 2. Given a stack with repeated elements, design an approach to compress it (value & frequency) while maintaining stack operations. Apply stack ADT modifications. Analyze trade-off between time and space
27	192511260	S DHARSHINI	1. A circular queue has a size of 6. Execute the following operations: Enqueue(10), Enqueue(20), Enqueue(30), Dequeue(), Enqueue(40), Enqueue(50), Enqueue(60), Enqueue(70). Draw the queue after each operation and identify the values of the front and rear pointers. 2. Design a system that dynamically chooses between linear search and binary search based on: ➤ Data size ➤ Sortedness ➤ Frequency of queries Analyze decision criteria mathematically
28	192471038	SASI KUMAR MEGHA	1. Insert the following elements 100, 200, 300, 400, 500 into the linked stack and pop the elements 100 and 200 from the linked stack. Show the linked structure before and after linked stack operations. 2. An online food delivery service receives the following orders: 01, 02, 03, 04. While 02 is being prepared, an express order 05 arrives. <u>Questions:</u> ➤ Explain why a normal queue may not satisfy customer requirements. ➤ Recommend an appropriate queue implementation and justify your choice.
29	192424359	SUNDARESAN R U	1. Construct an efficient system using two stacks within a single array structure to maximize memory utilization. Compare fixed partitioning and dynamic allocation strategies, and analyze how each approach affects overflow conditions, space efficiency, and operational complexity under varying workloads. 2. A queue-based ticketing system serves customers in the order of arrival, but VIP customers must be prioritized occasionally. Analyze how modifying a simple queue into a priority-based queue affects fairness, waiting time, and computational complexity. Compare both systems using analytical reasoning.
30	192524479	SHAHANA	1. Convert the following infix expression into postfix notation. Show the stack contents after processing each symbol. $((A+B)*C-(D-E)*(F+G))/H$ 2. For the expression. $((8+4)*(7-3)/(2+1))+(5*(6-2))$ Perform the following:

			- Convert the infix expression into postfix notation - Show the operator stack after every step - Evaluate the postfix expression using a stack - State the final result.
31	192525185	SHAIK ROSHNI	- Ten documents are requested for printing. Illustrate the mechanism of printer operations. Consider the document number as 1003, 1008, 1010, 1005, 1009, 1004. Identify the suitable data structure and draw the diagram to print the documents. - The university announced the selected candidates register number for placement training. The student XXX, Register Number 20142010 wishes to check whether his name is listed or not. The list is not sorted in any order. Identify the searching technique that can be applied and explain the searching steps with suitable procedure. List includes 20142015, 20142033, 20142011, 20142017, 20142010, 20142056, 20142003
32	192525214	SOPPA JASWANTH	- Insert the following elements 150, 250, 350, 450, 550 into the double linked list and delete the elements 150 and 250 from the double linked list. Show the linked structure using a node diagram before and after delete operations. - A circular queue has a size of 6. Execute the following operations: Enqueue(10), Enqueue(20), Enqueue(30), Dequeue(), Enqueue(40), Enqueue(50), Enqueue(60), Enqueue(70). Draw the queue after each operation and identify the values of the front and rear pointers.
33	192524023	SREE B	- Convert the expression $((a + b) + c * (d + e) + f) * (g + h)$ to 1) Prefix expression, 2) Postfix expression. - Perform the binary search in an array of n elements using a recursion function, in which a function has to call by itself. Use these elements given in sorted order and assume 88 as the key element. [23, 33, 43, 48, 53, 58, 61, 64, 68, 73, 88]
34	192521097	SUMETHA SRI S	How a queue may be maintained by a circular array with MAXSIZE = 6. Observe that queue always occupies consecutive locations except when it occupies locations at the beginning and at the end of the array. If the queue is viewed as a circular array, this means that it still occupies consecutive locations. The queue is empty when count = 0; Perform the following Enqueue and Dequeue operations. a) Initially QUEUE is empty, b) A, B, C are Enqueued, c) A is Dequeued, d) D, E, F are Enqueued, e) B & C are Dequeued, f) G is Enqueued, g) D & E are Dequeued, h) H & I are Enqueued, i) F is dequeued, j) G & H are dequeued k) I is Dequeued. Demonstrate with a neat sketch of Front, Rear and count in a queue and circular queue.

			2. Show the linked list configuration resulting from each series of list operations using the List ADT. Assume that lists L1 and L2 are empty at the beginning of each series. Show where the current position is in the list. (a) L1.append(10); L1.append(20); L1.append(15); (b) L2.append(10); L2.append(20); L2.append(15); L2.moveToStart(); L2.insert(39); L2.next(); L2.insert(12);
35	192465038	THAMMISETTI RAJU	1. Convert the following infix expression into postfix notation. $((A+B)*(C-(D/E))+F)*(G-H)$. Also determine the maximum stack size required during the conversion. 2. A circular parking system has 6 parking slots. Operations: ➤ Car1 enters ➤ Car2 enters ➤ Car3 enters ➤ Car1 exits ➤ Car4 enters ➤ Car5 enters ➤ Car6 enters ➤ Car7 enters Questions: ➤ Draw the circular queue after each operation. ➤ Identify the positions of the front and rear pointers. ➤ Explain why a circular queue is more efficient than a linear queue in this scenario.
36	192565092	THEJAS B	1. Given a singly linked list L: L1-> L2->L3...-> Ln-1->Ln, reorder and print it to: L1-> Ln ->L2 -> Ln-1. 2. Insert the following elements 150, 250, 350, 450, 550 into the double linked list and delete the elements 150 and 250 from the double linked list. Show the linked structure using a node diagram before and after delete operations.
37	192511217	THOTA VARUN TEJ	1. A circular queue has a size of 6. Execute the following operations: Enqueue(10), Enqueue(20), Enqueue(30), Dequeue(), Enqueue(40), Enqueue(50), Enqueue(60), Enqueue(70). Draw the queue after each operation and identify the values of the front and rear pointers. 2. A router processes packets using a queue. Packets arrive in the order: P1, P2, P3, P4, P5. The buffer size is 4 packets. Questions: ➤ Identify which packet is dropped. ➤ Explain why the packet is discarded. ➤ Suggest a better queue management strategy to reduce packet loss.

38	192524251	VAISHNAV M	1. Two algorithms solve the same problem. Algorithm A uses recursion, while Algorithm B uses an explicit stack. Compare both approaches in terms of memory usage, execution flow, and risk of stack overflow. 2. A stack has a capacity of 5 elements. Explain what happens if the following operations are executed: Push(A), Push(B), Push(C), Push(D), Push(E), Push(F). Identify the error and suggest two possible solutions.
39	192525187	VEERA SANTHOSH A	1. Compare the performance of a linear queue and a circular queue when 100 enqueue and dequeue operations are performed alternately. 2. A queue implemented using an array report "Queue Full" even though some positions are empty. ➤ Analyze why this occurs. ➤ Explain how a circular queue solves this problem.
40	192511226	VINIL KUMAR S	1. Consider the singly linked list: 10 → 20 → 30 → 40 → 50. Delete the node containing 30. Explain how the links change after deletion. 2. Given two sorted linked lists: L1: 1 → 3 → 5 → 7 L2: 2 → 4 → 6 → 8

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge	Course Coordinator	Program Director
Signature: 	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

convert the expression $((a+b) + c * (d+e) + f) * (g+h)$ to

D. Pre fix Expression

Step:1 Reverse the given expression

$$(h+g) * (f + (e+d) * c + (b+a))$$

Input	stack	output
(		hg+fed+c
h		hg+fed+c
+		hg+fed+c
g		hg+fed+c
)		hg+fed+c
+		hg+fed+c
(		hg+fed+c
f		hg+fed+c
+		hg+fed+c
(		hg+fed+c
e		hg+fed+c
+		hg+fed+c
d		hg+fed+c
)		hg+fed+c
*		hg+fed+c
c		hg+fed+c
+		hg+fed+c
(		hg+fed+c
b		hg+fed+c
+		hg+fed+c
a		hg+fed+c
)		hg+fed+c

The Pre-fix expression is

++ab*c-d e f+g+h

(ii) Post-fix expression

Input	Stack	Output
(		
(		
a		
+		
b		
)		
*		
c		
-		
(		
d		
e		
)		
+		
f		
+		
g		
+		
h		
*		

The post fix expression is ab+cd e f+g h+*

Recursive Binary Search

Given the sorted array:

[23, 33, 43, 48, 53, 58, 61, 64, 68, 73, 87]

Number of elements $(n) = 11$

Key element = 87

Algorithm

1. Find the middle element using

$$\text{mid} = \frac{\text{low} + \text{high}}{2}$$

2. Compare the key with middle element.

3. If the key equals the middle element, return its position.

4. If the key is greater than the middle element, recursively search the right half.

5. If the key is smaller than the middle element, recursively search the left half.

6. Repeat until the key is found or the search interval becomes empty.

1	2	3	4	5	6	7	8	9	10	11
23	33	43	48	53	58	61	64	68	73	87

↑
low

$$\text{mid} = \frac{23 + 87}{2}$$

↑
high

$$\text{mid} = \frac{110}{2} = 55$$

Step: 1

58 < 87

so = ~~high~~ = high - 1 = 11 - 1 = 10

Step: 2

1	2	3	4	5	6	7	8	9	10
23	33	43	48	53	58	61	64	68	73

↑
Low

↑
mid

↑
high

$$\text{mid} = \frac{73 + 23}{2} = \frac{96}{2} = 48$$

Step: 3

1	2	3	4	5	6	7	8	9	10	11
23	33	43	48	53	58	61	64	68	73	87

↑
low

↑
mid

↑
high

48 < 87,

$$\text{so } \text{low} = \text{low} + 1 = 1 + 1 = 2$$

$$\text{new mid} = \frac{33 + 73}{2} = \frac{106}{2} = 53$$

Step: 4

1	2	3	4	5	6	7	8	9	10	11
23	33	43	48	53	58	61	64	68	73	87

↑
low

↑
mid

↑
high

$$\therefore 53 < 87$$

$$\text{new low} = 2 + 1 = 3$$

$$\text{mid} = \frac{43 + 73}{2} = \frac{116}{2} = 58$$

Step: 5

1	2	3	4	5	6	7	8	9	10	11
23	33	43	48	53	58	61	64	68	73	87

↑
low

↑
mid

↑
high

	mid-1 Low	mid-1 High	Mid	middle element	Comparison	Next Recursive call
	0	10	5	58	87 > 58	Search Right (6-10)
2	6	10	8	68	87 > 68	Search Right (8-10)
3	9	10	9	73	87 > 73	Search Right (10-10)
4	10	10	10	87	8 = 87	Element Found

Recursive Function Calls

Binary Search (0, 10)

↓

Binary Search (6, 10)

↓

Binary Search (9, 10)

↓

Binary Search (10, 10)

↓

Element Found (87)

Result:

* Key Element: 87

* Position (0-based index): 10

* Position (1-based index): 11

* Total Recursive Calls: 4

Analysis

- * Since the array is sorted, Binary Search repeatedly divides the search space into two equal halves.
- * The recursion function calls itself on only one half of the array after each comparison.
- * The search continues until the key is found or no element remains.
- * In this example, the element 87 is found after four recursive calls, demonstrating the efficiency of binary search.

Time Complexity: $O(\log n)$

Space Complexity (Recursive): $O(\log n)$ due to the recursion call stack

Woj